

Kinematics

1 Introduction

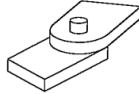
Kinematics is the science of motion that treats the subject without regard to the forces that cause it. Within the science of kinematics, one studies the position, the velocity, the acceleration, and all higher order derivatives of the position variables(with respect to time or any other variable(s)).

Links and joints are the fundamental components of a robot. **Joints** represent movement on one rotational axis(x, y or z as rotational axis), or a one parameter translation(motion on x,y, or z axis)(one degree of freedom per joint). To fully represent an object in 3d space, we need six parameters; three for rotation (α, β, γ) and three for translation((x, y, z) -position of joint) (any more is considered redundant), because of this most robots are constructed using six joints to be able to fully express their position and orientation in 3d. **Links** are the static(unchanging) parts of a robot that connect joints to each other. Given n joints you will have n-1 links; think of links as edges and joints as nodes in a graph.

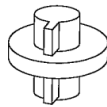
Examples of joints:

prismatic, spherical, planar(translation with one degree of freedom)

revolute, screw, cylindrical(rotational with one degree of freedom)



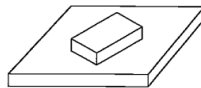
Revolute



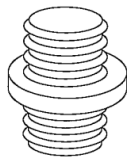
Prismatic



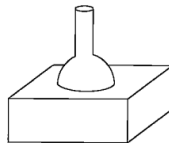
Cylindrical



Planar

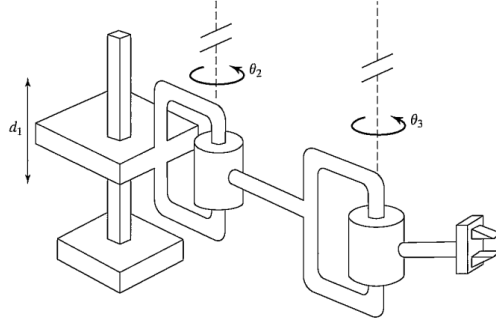


Screw



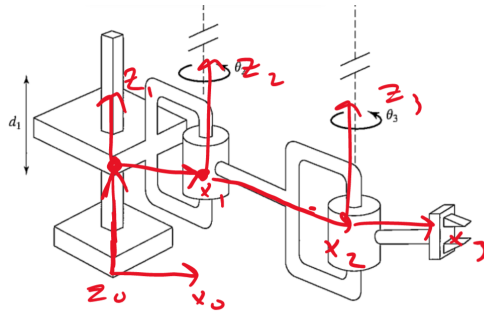
Spherical

2 Conventions for link-frame assignment



Algorithm:

1. Identify the joint axes and imagine (or draw) infinite lines along them. For steps 2 through 5 below, consider two of these neighboring lines (at axes i and $i + 1$).
2. Identify the common perpendicular between them, or point of intersection. At the point of intersection, or at the point where the common perpendicular meets the i th axis, assign the link-frame origin.
3. Assign the z_i axis pointing along the i th joint axis.
4. Assign the x_i axis pointing along the common perpendicular, or, if the axes intersect, assign x_i to be normal to the plane containing the two axes.
5. Assign the y_i axis to complete a right-hand coordinate system (optional step).
6. Assign frame $\{0\}$ to match frame $\{1\}$ when the first joint variable is zero. For N , choose an origin location and x_n direction freely, but generally so as to cause as many linkage parameters as possible to become zero.



3 Parameters for dh table

The DH table is a method for organizing parameters from a system of known joints, links, and frames. These parameters give us general formulas for constructing the transformations matrices required for computations.

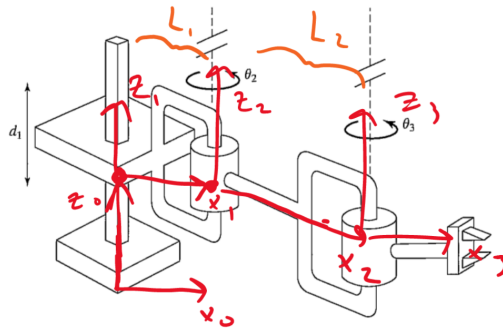
a_i =Distance from z_i to z_{i+1} along the x_i (this is the link length, and mutual perpendicular between frame $\{i\}$ and $\{i+1\}$)

α_i =link twist angle, angle from z_i to z_{i+1} measured along x_i (angle between axis i and $i+1$)

d_i =link offset, the distance from x_{i-1} to x_i measured along z_i (the offset distance of one link to the next)

θ_i =the angle from x_{i-1} to x_i measured along z_i (rotation of link w.r.t. its neighbor along common axis)

DH table				
Frame $\{i\}$	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	d_1	$\theta_1 = 0$
2	L_1	0	0	θ_2
3	L_2	0	0	θ_3



4 dh table to transformation matrix

Now that we have filled out the DH table, we can make all our transformation matrices using the matrix formula, Note the transformation uses rotation basis: $(R_z(\theta_i)R_x(\alpha_{i-1}))^T$. Tip - Each row corresponds to one transformation matrix, so row 1 has all the variables need for 0_1T , row 2 has all the variables for 1_2T and so on.

$${}_{i-1}^iT = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Note: Use $c0=1$ and $s0=0$ to simplify.

$${}^0_1T = \begin{bmatrix} c0 & -s0 & 0 & a_0 \\ s0c0 & c0c0 & -s0 & -s0d_1 \\ s0c0 & s0c0 & c0 & c0d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

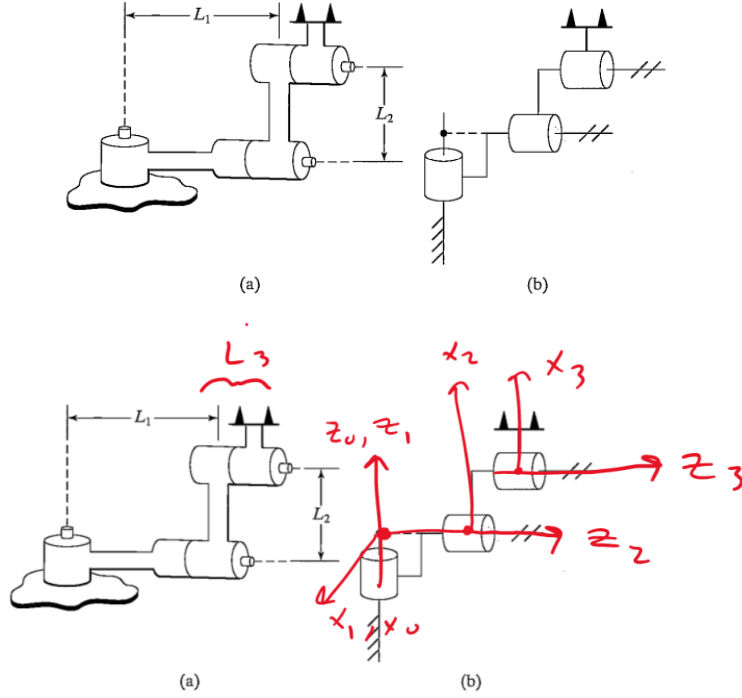
$${}^1_2T = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & L_1 \\ s\theta_2c0 & c\theta_2c0 & -s0 & -s0d_2 \\ s0c0 & s0c0 & c0 & c0d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & L_1 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & L_2 \\ s\theta_3c0 & c\theta_3c0 & -s0 & -s0d_3 \\ s0c0 & s0c0 & c0 & c0d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & L_2 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The whole system can be manipulated using the transformation:

$${}^0_3T = {}^0_1T {}^1_2T {}^2_3T = {}^0_3T = \begin{bmatrix} \cos(\theta_2 + \theta_3) & -\sin(\theta_2 + \theta_3) & 0 & L_1 + L_2 \cos \theta_2 \\ \sin(\theta_2 + \theta_3) & \cos(\theta_2 + \theta_3) & 0 & L_2 \sin \theta_2 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

5 2nd Example



$$\begin{aligned}
 a_0 &= \text{dist}(z_0, z_1) \text{ in } x_0 = 0 \\
 a_1 &= \text{dist}(z_1, z_2) \text{ in } x_1 = 0 \\
 a_2 &= \text{dist}(z_2, z_3) \text{ in } x_2 = L_2
 \end{aligned}$$

$$\begin{aligned}
 \alpha_0 &= \text{angle}(z_0, z_1) \text{ in } x_0 = 0 \\
 \alpha_1 &= \text{angle}(z_1, z_2) \text{ in } x_1 = -90^\circ \\
 \alpha_2 &= \text{angle}(z_2, z_3) \text{ in } x_2 = 0
 \end{aligned}$$

$$\begin{aligned}
 d_1 &= \text{dist}(x_0, x_1) \text{ in } z_1 = 0 \\
 d_2 &= \text{dist}(x_1, x_2) \text{ in } z_2 = L_1 \\
 d_3 &= \text{dist}(x_2, x_3) \text{ in } z_3 = L_3
 \end{aligned}$$

$$\begin{aligned}
 \theta_1 &= \text{angle}(x_0, x_1) \text{ in } z_1 = \theta_1 \\
 \theta_2 &= \text{angle}(x_1, x_2) \text{ in } z_2 = \theta_2 - 90^\circ \\
 \theta_3 &= \text{angle}(x_2, x_3) \text{ in } z_3 = \theta_3
 \end{aligned}$$

DH table				
Frame {i}	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	-90°	L_1	$\theta_2 - 90^\circ$
3	L_2	0	L_3	θ_3

Using ${}^{i-1}_i T = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$ we get:

$${}^0_1 T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2 T = \begin{bmatrix} c(\theta_2 - 90^\circ) & -s(\theta_2 - 90^\circ) & 0 & 0 \\ 0 & 0 & 1 & L_1 \\ s(\theta_2 - 90^\circ) & c(\theta_2 - 90^\circ) & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3 T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & L_2 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & L_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Ultimately we get:

$${}^0_3 T = \begin{bmatrix} c_1 s_{23} & c_1 c_{23} & -s_1 & L_2 c_1 s_2 - (L_1 + L_3) s_1 \\ s_1 s_{23} & s_1 c_{23} & c_1 & L_2 s_1 s_2 + (L_1 + L_3) c_1 \\ c_{23} & -s_{23} & 0 & L_2 c_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where:

$$c_1 = \cos \theta_1, \quad s_1 = \sin \theta_1$$

$$c_2 = \cos \theta_2, \quad s_2 = \sin \theta_2$$

$$c_{23} = \cos(\theta_2 + \theta_3), \quad s_{23} = \sin(\theta_2 + \theta_3)$$

6 Jacobian Matrix and Singularities

Definition of Jacobian - multidimensional matrix of derivatives (J → velocity differentiate from positions), Jacobians can relate the joint velocities to cartesian(end effector) velocities in a robot manipulator.

$$\text{Cartesian: } v = \begin{bmatrix} \dot{P}_x \\ \dot{P}_y \\ \dot{P}_z \end{bmatrix} = \text{Joint: } \begin{bmatrix} {}^0v \\ {}^0w \end{bmatrix} = J * \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

0v is linear

0w is angular

You always need the jacobian in the base frame 0J , when you don't have it; you can solve for it. Use formula:

$${}^0J = {}^0R * {}^xJ$$

Singularities- is a point where the robot loses one or more degrees of freedom. Here, it is impossible to move the end effector (tool) in a particular direction, regardless of the joint rates. two types:

- Workspace-bound singularities- fully stretched out , or tool folded back on itself. Tool is at boundary of workspace.
- Workspace interior singularities - 2 or more joint axes line up with each other.

- At singularities robots lose their degrees of freedom. if the $\det[J(\theta)] = 0$ singularity exist, if $\det[J(\theta)] \neq 0$ no singularities. Remember $\theta = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \dots \\ \theta_n \end{bmatrix}$

7 Jacobian Matrix -(to get robot velocities)

Jacobian involves finding velocities. Steps:

- DH table $\rightarrow$ Transformation matrix
- Get ${}^0T = {}^0T_1 {}^1T_2 \dots {}^nT_e$ and extract translation column to get $P = \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix}$ (position of end effector relative to base frame)
- When we differentiate P, we get: $\begin{bmatrix} \dot{P}_x \\ \dot{P}_y \\ \dot{P}_z \end{bmatrix} = {}^0J \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix}$

General formula:

Given position of joints:

- $y_1 = f_1(x_1, x_2, \dots)$
- $y_2 = f_2(x_1, x_2, \dots)$

- ...
- $y_n = f_n(x_1, x_2, \dots)$

We can then differentiate to get:

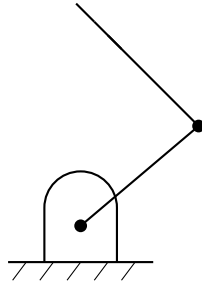
- $\partial y_1 = f_1(\frac{\partial f_1}{\partial x_1} \partial x_1, \frac{\partial f_1}{\partial x_2} \partial x_2, \dots)$
- $\partial y_2 = f_2(\frac{\partial f_2}{\partial x_1} \partial x_1, \frac{\partial f_2}{\partial x_2} \partial x_2, \dots)$
- ...
- $\partial y_n = f_n(\frac{\partial f_n}{\partial x_1} \partial x_1, \frac{\partial f_n}{\partial x_2} \partial x_2, \dots)$

In Matrix form: $\dot{Y} = J\dot{X} \rightarrow \begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \\ \dots \end{bmatrix} = \left[\frac{\partial F}{\partial X} = J \right] \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dots \end{bmatrix}$

7.1 Example

Q: Velocity of the tip of the arm as a function of joint rates

- Give the answer in terms of frame 0
- Give the answer in terms of frame 2



Axis (i)	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	L_1	0	θ_1
2	0	L_2	0	θ_2

Part 1 using kinematics formula to get:

$${}^0_1T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & l_1 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2T = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & l_2 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Now we can solve for:

$${}^0_2T = \begin{bmatrix} c_{12} & -s_{12} & 0 & l_1 + l_2 c\theta_1 \\ s_{12} & c_{12} & 0 & l_2 s\theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Giving us : $P = \begin{bmatrix} l_1 + l_2 c\theta_1 \\ l_2 s\theta_1 \\ 0 \end{bmatrix}$ thus, : $\dot{P} = \begin{bmatrix} -l_2 s\theta_1 \dot{\theta}_1 \\ l_2 c\theta_1 \dot{\theta}_1 \\ 0 \end{bmatrix}$ We can then write:

$$\dot{P} = \begin{bmatrix} -l_2 s\theta_1 \\ l_2 c\theta_1 \\ 0 \end{bmatrix} [\dot{\theta}_1] \text{ where this is equivalent to } \dot{Y} = J\dot{X} \text{ and } {}^0V = {}^0J[\dot{\theta}_1]$$

ANSWERS QUESTION VELOCITY OF THE TOOL(FRAME 2) RELATIVE TO FRAME 0

Part 2

Solve ${}^2V = {}^2J[\dot{\theta}_1]$. Using formula:

$${}^2J = {}^2_0R^0J$$

From previous question, where we solved for 0_2T we can extract 0_2R and take transpose to get:

$${}^2_0R = \begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ so, we get:}$$

$${}^2J = \begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -l_2 s\theta_1 \\ l_2 c\theta_1 \\ 0 \end{bmatrix} = \begin{bmatrix} -l_2 c_{12} s_1 + l_2 s_{12} c_1 \\ l_2 s_{12} s_1 + l_2 c_{12} c_1 \\ 0 \end{bmatrix}$$

and finally solving for 2v , we get:

$${}^2v = \begin{bmatrix} -l_2 c_{12} s_1 + l_2 s_{12} c_1 \\ l_2 s_{12} s_1 + l_2 c_{12} c_1 \\ 0 \end{bmatrix} [\dot{\theta}_1]$$

8 Jacobian Matrix - Velocity Propagation Method

Intro Steps:

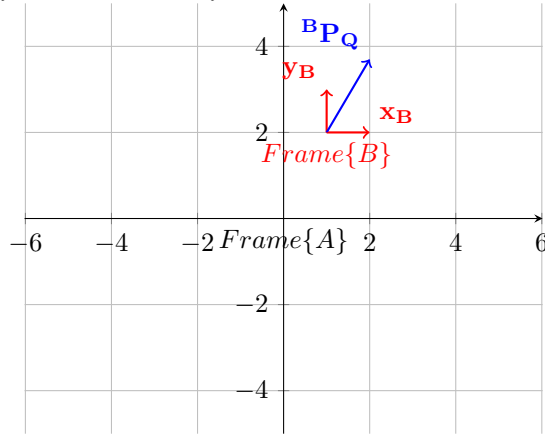
- Get linear velocity (v)

- Get angular velocity (ω)
- Start from first frame and solve eqns till you get to end effector frame
 $0 \rightarrow 1 \rightarrow 2 \rightarrow \dots \rightarrow E$
- ${}^E V_E = {}^E J \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$
 ${}^0 J = {}^0_E R * {}^E J$

Linear Velocity

Consider two frames $\{A\}, \{B\}$; where $\{A\}$ is fixed and $\{B\}$ is moving away from $\{A\}$. Orientation of $\{B\}$ does not change. Let Frame $\{B\}$ have a point $Q({}^B P_Q)$. We can compute the linear velocity of ${}^B P_Q$ with respect to frame $\{A\}$ using the formula:

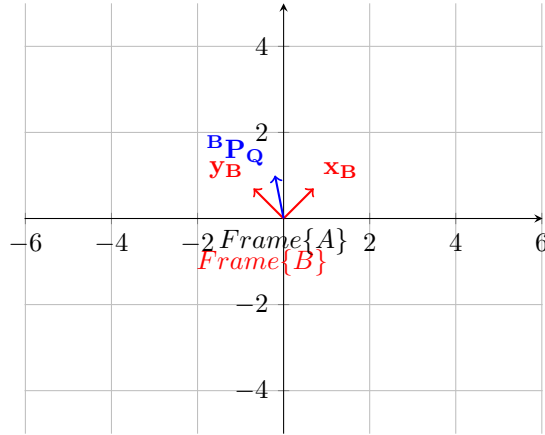
$${}^A V_Q = {}^A V_B + {}^A R^B V_Q$$



Angular Velocity

Consider two frames $\{A\}, \{B\}$; where $\{A\}$ is fixed and $\{B\}$ is moving away from $\{A\}$. No linear motion for $\{B\}$. Let Frame $\{B\}$ have a point $Q({}^B P_Q)$. We can compute the angular velocity of ${}^B P_Q$ with respect to frame $\{A\}$ using the formula:

$${}^A V_Q = {}^A R^B V_Q + {}^A \Omega \times {}^A R^B P_Q$$



8.1 Velocity Propagation equations(iterative)

1. ${}^{i+1}\omega_{i+1} = {}^i R({}^i\omega_i) + \dot{\theta}_{i+1} {}^{i+1}\hat{z}_{i+1}$
2. ${}^{i+1}v_{i+1} = {}^i R({}^i v_i + {}^i\omega_i \times {}^i P_{i+1}) + \dot{d}_{i+1} {}^{i+1}\hat{z}_{i+1}$

- i is joint/frame
- $i+1$ next frame
- ω angular velocity
- v linear velocity
- $\dot{\theta}_{i+1}\hat{z}$ is for revolute joints: $\begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_{i+1} \end{bmatrix}$
- $\dot{d}_{i+1}\hat{z}$ is for prismatic joints: $\begin{bmatrix} 0 \\ 0 \\ \dot{d}_{i+1} \end{bmatrix}$
- The main idea is starting from frame $\{0\}$ iteratively solve the solution for n joints.

8.2 Example

- (a) Find the **linear and angular velocity** at the last frame
- (b) Find the **Jacobian** at the last frame
- (c) Find the **Jacobian** at the base frame
- (d) Find an equation for the **singularities** of the robot

The solution procedure is:

1. DH Table
2. Transformation matrices
3. Velocity propagation: base $\rightarrow$ final
4. 3J
5. 0J
6. Singularities

Define

$$c_i = \cos \theta_i, \quad s_i = \sin \theta_i.$$

The individual transformation matrices are

$${}^0_1T = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$${}^1_2T = \begin{bmatrix} -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ c_2 & -s_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

and

$${}^2_3T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -d_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

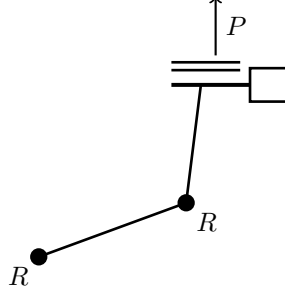
Multiplying the transformations gives

$${}^0_3T = {}^0_1T {}^1_2T {}^2_3T,$$

so that

$${}^0_3T = \begin{bmatrix} -c_1s_2 & s_1 & c_1c_2 & c_1c_2d_3 \\ -s_1s_2 & c_1 & s_1c_2 & s_1c_2d_3 \\ c_2 & 0 & s_2 & s_2d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Robot Configuration



Thus, the robot has the joint sequence

$$\boxed{R - R - P}$$

where the first two joints are revolute and the third joint is prismatic.

8.2.1 Part A(solution)

Base frame usually never moves so we can say: **frame{0}**

$${}^0\omega_0 = {}^0v_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Frame {1} iteration(i=0):

$$1. {}^1\omega_1 = {}^1_0R({}^0\omega_0) + \dot{\theta}_1 \hat{z} = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix}$$

$$2. {}^1v_1 = {}^1_0R({}^0v_0 + {}^0\omega_0 \times {}^0P_1) + d_1 {}^1\dot{z}_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad d_1 {}^1\dot{z}_1 \text{ goes to zero cause revolute joint here.}$$

Frame {2} iteration(i=1):

$$1. {}^2\omega_2 = {}^2_1R({}^1\omega_1) + \dot{\theta}_2 \hat{z} = \begin{bmatrix} -s_2 & 0 & c_2 \\ -c_2 & 0 & -s_2 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_2 \end{bmatrix} = \begin{bmatrix} c_2 \dot{\theta}_1 \\ -s_2 \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

$$2. {}^2v_2 = {}^2_1R({}^1v_1 + {}^1\omega_1 \times {}^1P_2) + d_2 {}^2\dot{z}_2 = \begin{bmatrix} -s_2 & 0 & c_2 \\ -c_2 & 0 & -s_2 \\ 0 & -1 & 0 \end{bmatrix} \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right) =$$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad d_2 {}^2\dot{z}_2 \text{ goes to zero cause revolute joint here.}$$

Frame {3} iteration(i=2):

$$1. {}^3\omega_3 = {}^3R({}^2\omega_2) + \dot{\theta}_3 \hat{z} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} c_2 \dot{\theta}_1 \\ -s_2 \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} = \begin{bmatrix} c_2 \dot{\theta}_1 \\ \dot{\theta}_2 \\ s_2 \dot{\theta}_1 \end{bmatrix} \quad \dot{\theta}_2^2 \hat{z}_2 \text{ goes to zero}$$

cause prismatic joint here.

$$2. {}^3v_3 = {}^3R({}^2v_2 + {}^2\omega_2 \times {}^2P_3) + \dot{d}_3 {}^3\hat{z}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} c_2 \dot{\theta}_1 \\ -s_2 \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \times \begin{bmatrix} 0 \\ d_3 \\ 0 \end{bmatrix} \right) +$$

$$\begin{bmatrix} 0 \\ 0 \\ \dot{d}_3 \end{bmatrix} = \begin{bmatrix} d_3 \dot{\theta}_2 \\ -d_3 c_2 \dot{\theta}_1 \\ \dot{d}_3 \end{bmatrix}$$

8.2.2 Part B(solution)

Find 3J . Note when solving for jacobian, you only need to use linear velocity.

$${}^3v = \begin{bmatrix} d_3 \dot{\theta}_2 \\ -d_3 c_2 \dot{\theta}_1 \\ \dot{d}_3 \end{bmatrix} = \begin{bmatrix} 0 & d_3 & 0 \\ -d_3 c_2 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{d}_3 \end{bmatrix} \rightarrow \dot{Y} = J\dot{X}$$

8.2.3 Part C(solution)

Find 0J .

$${}^0J = {}^0R^3J = \begin{bmatrix} -c_{12} & s_1 & c_1 c_2 \\ -s_1 s_2 & c_1 & s_1 c_2 \\ c_2 & 0 & s_2 \end{bmatrix} \begin{bmatrix} 0 & d_3 & 0 \\ -d_3 c_2 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -s_1 d_3 c_2 & -c_1 s_2 d_3 & c_1 c_2 \\ -c_1 d_3 c_2 & -s_1 s_2 d_3 & s_1 c_2 \\ 0 & c_2 d_3 & s_2 \end{bmatrix}$$

8.2.4 Part D(solution)

Find singularities by taking determinant: You must solve the unknowns in $\det[{}^0J]$. You can use levitas determinant or some software. The unknowns will be θ_1, θ_2, d_3 and these are values for parameters that reduce DOF of robot(aka robot must avoid these values).

9 Inverse Manipulator Kinematics

Kinematics is the science of motion that treats the subject without regard to the forces that cause it. Within the science of kinematics, one studies the position, the velocity, the acceleration, and all higher order derivatives of the position variables (with respect to time or any other variable(s)). Hence, the study of the kinematics of manipulators refers to all the geometrical and time-based properties of the motion.

Inverse Kinematics asks: Given the desired position and orientation of the tool relative to the station, how do we compute the set of joint angles which will achieve this desired result?

Solving the problem of finding the required joint angles to place the tool frame, $\{T\}$, relative to the station frame, $\{S\}$, is split into two parts. First, frame transformations are performed to find the wrist frame, $\{W\}$, relative to the base frame, $\{B\}$, and then the inverse kinematics are used to solve for the joint angles.

10 Computing Inverse Kinematics

Consider a planar 2R robot with link lengths L_1 and L_2 . The joint variables are

$$q = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}.$$

The forward kinematics for the end-effector position are

$$x = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)$$

$$y = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2).$$

Thus,

$$P(q) = \begin{bmatrix} L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2) \\ L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2) \end{bmatrix}.$$

Given a desired end-effector position

$$P_d = \begin{bmatrix} x_d \\ y_d \end{bmatrix},$$

the inverse kinematics problem consists of finding the joint variables θ_1 and θ_2 that satisfy

$$P(q) = P_d.$$

10.1 Analytical Inverse Kinematics

Starting from the forward kinematics equations,

$$x_d = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)$$

$$y_d = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2),$$

we first solve for θ_2 .

Using the law of cosines,

$$x_d^2 + y_d^2 = L_1^2 + L_2^2 + 2L_1L_2 \cos \theta_2.$$

Therefore,

$$\cos \theta_2 = \frac{x_d^2 + y_d^2 - L_1^2 - L_2^2}{2L_1L_2}$$

and

$$\sin \theta_2 = \pm \sqrt{1 - \cos^2 \theta_2}.$$

Thus,

$$\theta_2 = \text{atan2} \left(\pm \sqrt{1 - \cos^2 \theta_2}, \cos \theta_2 \right)$$

where the two signs correspond to the two possible configurations, commonly called elbow-up and elbow-down.

Once θ_2 is known, θ_1 can be calculated as

$$\theta_1 = \text{atan2}(y_d, x_d) - \text{atan2}(L_2 \sin \theta_2, L_1 + L_2 \cos \theta_2)$$

Therefore, the analytical inverse kinematics solution is

$$q = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}$$

with θ_1 and θ_2 given by the equations above.

10.2 Analytical IK Example

Let

$$L_1 = L_2 = 1$$

and let the desired end-effector position be

$$P_d = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Solving for θ_2 ,

$$\cos \theta_2 = \frac{1^2 + 1^2 - 1^2 - 1^2}{2(1)(1)} = 0.$$

Therefore,

$$\theta_2 = \text{atan2}(1, 0) = 90^\circ.$$

Then,

$$\theta_1 = \text{atan2}(1, 1) - \text{atan2}(1, 1)$$

giving

$$\theta_1 = 0^\circ.$$

Thus, one inverse kinematics solution is

$$q = \begin{bmatrix} 0^\circ \\ 90^\circ \end{bmatrix}.$$

10.3 Numerical Inverse Kinematics

Inverse kinematics can also be solved numerically using the Jacobian.

For small changes in joint position,

$$\Delta P \approx J(q)\Delta q.$$

Therefore, if the desired change in end-effector position is known,

$$\Delta q \approx J^{-1}(q)\Delta P.$$

Define the position error at iteration k as

$$e_k = P_d - P(q_k)$$

where q_k is the current estimate of the joint angles.

The joint correction is then

$$\Delta q_k = J^{-1}(q_k)e_k$$

and the new joint estimate is

$$q_{k+1} = q_k + \Delta q_k.$$

Therefore,

$$q_{k+1} = q_k + J^{-1}(q_k) [P_d - P(q_k)]$$

This process is repeated until the position error is sufficiently small:

$$\|P_d - P(q_k)\| < \epsilon.$$

10.4 Jacobian for the Planar 2R Robot

The position is

$$P(q) = \begin{bmatrix} x(q) \\ y(q) \end{bmatrix}.$$

The Jacobian is

$$J(q) = \frac{\partial P}{\partial q} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} \end{bmatrix}.$$

Taking the partial derivatives gives

$$J(q) = \begin{bmatrix} -L_1 \sin \theta_1 - L_2 \sin(\theta_1 + \theta_2) & -L_2 \sin(\theta_1 + \theta_2) \\ L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2) & L_2 \cos(\theta_1 + \theta_2) \end{bmatrix}.$$

10.5 Numerical IK Example

Again, let

$$L_1 = L_2 = 1$$

with desired position

$$P_d = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Choose the initial guess

$$q_0 = \begin{bmatrix} 30^\circ \\ 30^\circ \end{bmatrix} = \begin{bmatrix} 0.524 \\ 0.524 \end{bmatrix} \text{ rad.}$$

Using forward kinematics,

$$P(q_0) = \begin{bmatrix} \cos 30^\circ + \cos 60^\circ \\ \sin 30^\circ + \sin 60^\circ \end{bmatrix} = \begin{bmatrix} 1.366 \\ 1.366 \end{bmatrix}.$$

The error is therefore

$$e_0 = P_d - P(q_0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1.366 \\ 1.366 \end{bmatrix} = \begin{bmatrix} -0.366 \\ -0.366 \end{bmatrix}.$$

At the initial configuration,

$$J(q_0) = \begin{bmatrix} -1.366 & -0.866 \\ 1.366 & 0.5 \end{bmatrix}.$$

The first joint correction is

$$\Delta q_0 = J^{-1}(q_0)e_0 \approx \begin{bmatrix} -0.268 \\ 0.845 \end{bmatrix}.$$

Therefore,

$$q_1 = q_0 + \Delta q_0$$

gives

$$q_1 \approx \begin{bmatrix} 0.256 \\ 1.369 \end{bmatrix} \text{ rad}$$

or approximately

$$q_1 \approx \begin{bmatrix} 14.7^\circ \\ 78.4^\circ \end{bmatrix}.$$

The process is then repeated:

$$P(q_1) \rightarrow e_1 \rightarrow J(q_1) \rightarrow \Delta q_1 \rightarrow q_2.$$

After several iterations, the solution converges approximately to

$$q^* = \begin{bmatrix} 0^\circ \\ 90^\circ \end{bmatrix}.$$

10.6 Jacobian Pseudoinverse

For a general robot, the Jacobian may not be square or invertible. Therefore, numerical inverse kinematics commonly uses the Moore–Penrose pseudoinverse $J^\dagger$:

$$q_{k+1} = q_k + J^\dagger(q_k) [P_d - P(q_k)].$$

A step-size parameter α may also be introduced:

$$q_{k+1} = q_k + \alpha J^\dagger(q_k) [P_d - P(q_k)], \quad 0 < \alpha \leq 1.$$

The iterations continue until

$$\|P_d - P(q_k)\| < \epsilon.$$

11 Practice:

